Physics 610
Fall 2023
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Homework 4

Due Wednesday, October 4

1. Problem 17 to Chap. 5 of the textbook. Please treat this problem as rotation about a fixed point and go through the following steps.
   (a) Euler's angles are defined in Sec. 4.4 of the textbook, and the expression for the angular velocity $\boldsymbol{\omega}$ in terms of these angles and their derivatives is given in Problem 15 to Chap. 4 there. Use these expressions to obtain Euler's angles (θ, ϕ, ψ) as functions of time. Please be careful to distinguish the angular velocity ω from the angular frequency corresponding to the period τ of the motion. Please choose the axis of the cone as the z' axis of the rotating frame. You may assume that at time $t = 0$ this axis was in the yz plane of the lab frame.
   (b) Components of the angular velocity in the rotating frame are given in Eq. (4.87) of the textbook. Use your results from part (a) to find all three of these components.
   (c) Using suitable moments of inertia, find the kinetic energy and the vector of angular momentum of the cone about the vertex.

2. Problem 30 to Chap. 5 of the textbook.